# Where to Measure Next? Evaluating Feedback-Driven Region Selection for Spatial Transcriptomics

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## Abstract

Spatial transcriptomics is costly, so many studies profile only selected regions of a section, and recent methods choose each next region from expression already measured. Benchmarks of active spot selection for training predictors across patients report inconsistent gains over random sampling; we ask instead whether, within a slide, measured feedback improves selection over a fixed design matched to the objective. We built a retrospective simulator from 22 human HEST-1k slides (18 studies, four tissues), fixed the image encoder and imputer, registered three analysis stages in advance, and drew inference at the study level. **RQ1:** for whole-section reconstruction, a feedback policy targeting the imputer's error was worse than a non-adaptive design chosen on held-out studies at every budget from 0.5% to 20%, by up to 0.092 Pearson correlation; a semi-synthetic control did not establish sensitivity to reconstruction gains. **RQ2:** for capturing heterogeneous regions, informative feedback raised recall over permuted feedback by up to 0.063, but no margin survived multiplicity correction, and a diversity sampler did about as well. **RQ3:** when a coverage design spent half or more of the budget and feedback the rest, no hybrid beat the coverage design; from 2% upward all were within 0.007 correlation and 0.006 recall, although gains near 0.02 and 0.03 would probably have been detected at 2–10%. Feedback was informative (permuted values made the same policies worse) but mainly offset the cost of leaving coverage. **RQ4** (exploratory): batches of five neither eroded nor established a detection advantage. Method developers should compare adaptive selection with objective-matched fixed designs and with itself fed uninformative values. For experimentalists, within simulated spot-level budgets, one imputer, and computational endpoints, no tested feedback policy outperformed a well-chosen fixed design.

## 1 Introduction

Spatial transcriptomics measures gene expression without dissociating tissue, so expression can be read in its histological context [30, 35, 39]. Whole-transcriptome assays remain costly, and region-of-interest platforms such as GeoMx DSP [29] profile only selected regions of a section. Expression elsewhere can be predicted from the co-registered hematoxylin and eosin (H&E) image [21, 43]. Which regions are measured therefore determines both how well the unmeasured tissue can be reconstructed and whether distinct tissue compartments are measured at all.

Choosing regions is an experimental-design problem, and recent methods solve it sequentially. SOFisher [26] uses reinforcement learning to move fields of view toward regions enriched for a target signal, and SCR$^2$-ST [47] learns sampling locations for expression prediction. S2-omics [46] takes the opposite approach and selects regions from histology alone, before any measurement. The appeal of sequential selection follows active learning [38]: each measurement can inform the next.

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Its cost is a measurement turnaround between choices. A recent benchmark of active spot selection for training expression predictors found inconsistent gains over random sampling at small budgets [50], but addressed a different decision (Section 2). We are not aware of an evaluation that compares feedback-driven region selection within a slide against fixed designs matched to the objective while ablating the measured feedback itself.

Theory gives different expectations for different goals. For reconstructing a Gaussian-process field with a known kernel, posterior variance depends only on where measurements are taken, so a design fixed in advance is as good as an adaptive one [24]; adaptive selection can gain only when the model's uncertainty depends on the observed values [15]. Imputers learned from images are not Gaussian processes with known kernels, so for reconstruction the question is empirical. Finding regions defined by their expression is closer to active search, where the targets can be located only by measuring and feedback is essential [13].

An evaluation that can separate these cases needs four features. The baseline must be a design that could be chosen before the target slide is profiled, because selecting the best design on the evaluation data biases the comparison [6, 41]. Generalisation must be assessed across studies, not within them [3], with slides from one study treated as dependent [12]. The contribution of feedback must be isolated by giving the same policy uninformative values, in the spirit of permutation tests for learning algorithms [32]. Finally, null results need positive controls and power analysis to be interpretable [1].

We built such an evaluation and asked four questions. **RQ1**: does conditioning region selection on measured expression improve whole-section reconstruction over a feasible non-adaptive design? **RQ2**: does it improve the capture of molecularly heterogeneous regions? **RQ3**: does feedback add to a strong coverage design when it allocates only the remainder of the budget? **RQ4**, an exploratory stress test: does a detection advantage persist when measurements arrive in batches? The study contributes an evaluation, not a method: a reproducible simulator on 22 human slides from 18 studies, analyses registered before each experimental stage, and evidence that the tested feedback policies offered no supported advantage over well-chosen fixed designs for either objective.

## 2 Related work

**Expression from histology.** Expression has been predicted from H&E by contrastive retrieval [43], flow matching over spatial neighbourhoods [20], diffusion [48], and autoregressive generation [33], and benchmarked on HEST-1k [21]. These methods take the measured regions as given. We hold the predictor fixed and vary only the regions it receives.

**Choosing what to measure.** Pool-based active learning selects queries by uncertainty [25], expected error reduction [36], or coverage of the input space [37]. Sensor placement offers the matching non-adaptive designs: variance- and mutual-information-greedy selection, which are near-optimal for Gaussian-process reconstruction [24]. Adaptive submodularity characterises when sequential choices can beat a fixed set [15], and active search addresses the related goal of collecting positives [13]. Batch acquisition reduces the information available to each choice [23], and global optimisation commonly begins with a space-filling design before switching to adaptive queries [22]. Active feature acquisition casts measurement as a sequential policy [45].

**Design in spatial omics.** Gene-panel selectors fix which genes to measure [9, 28]. Region selectors range from one-shot image rankers [44] and histology-based heterogeneity selection [46] to sequential reinforcement-learning samplers [26, 47]. Our evaluation includes reimplementations of these objectives and a native run of SOFisher. The closest evaluation is a retrospective benchmark of active spot selection on breast and skin cancer cohorts [50], which compared uncertainty- and diversity-based strategies with random sampling and found that active strategies did not consistently win at small budgets. Its question differs from ours in three ways. It selects spots to train a morphology-to-expression predictor that is evaluated on held-out patients, whereas we select regions within a slide for a fixed imputer. Its baseline is random sampling, whereas ours are non-adaptive designs chosen for each objective. It compares selection strategies, whereas we hold each strategy fixed and remove the information in the measured values.

# 3 Methods

## 3.1 Acquisition simulator

A slide is a set of regions, each with an image embedding, spatial coordinates, and measured expression. A policy reveals regions one at a time until a budget, a fixed fraction of the slide's regions, is spent; each revealed region exposes its measured expression to the policy. The simulator is retrospective, as in pool-based active-learning evaluation [38]: every region has been measured, so any selection can be scored against ground truth. For RQ4, regions are revealed in rounds, and within a round the policy can use only measurements available at its start. Stage 1 also tested contiguous blocks of 7 regions around a chosen seed, approximating field-of-view acquisition. Budgets range from 0.5% of regions (9 to 45 regions depending on slide size, comparable to ROI-based profiling) to 20%.

## 3.2 Slides and preprocessing

We used 22 human HEST-1k slides [21] from prostate (6), kidney (4), lung (6), and colon (6), of which 7 are Visium HD [10] and 15 standard Visium [39] (Appendix A). The tissues are those used to evaluate the one-shot ranker we compare against [44]. Selection rules were fixed before any result on these slides: human tissue, whole-transcriptome assay, at least 1,000 spots under tissue, a source image of at most 0.6 $\mu$m/px so that image patches are not substantially upsampled, one slide per specimen, and at most two slides per study. The slides come from 18 studies, which serve as the independent units of inference.

A region is one native spot, the unit at which HEST provides both expression and an image patch. Visium spots are 55 $\mu$m wide on a 100 $\mu$m pitch; the HD pseudo-spots distributed by HEST have a median spacing of 128 $\mu$m on our slides. Slides contain 1,766 to 9,080 regions. Each region's image input is the HEST 224 px patch at 0.5 $\mu$m/px, embedded with the frozen pathology foundation model UNI2-h [7]. Following the HEST benchmark, expression is log1p-transformed and genes detected in fewer than 10% of regions are removed [21].

## 3.3 Objectives

*Reconstruction* (RQ1, RQ3): a fixed imputer predicts the unprofiled regions, scored by the per-gene Pearson correlation over those regions averaged over the 50 most variable genes (PCC), the HEST benchmark target [21]. *Detection* (RQ2–RQ4): regions of interest (ROIs) are the 20% of regions with the highest local molecular heterogeneity, the variance of top-50-gene expression across each region's 8 nearest spatial neighbours, after the heterogeneity objective of S2-omics [46]. The ROI fraction equals the largest budget, so a perfect policy could capture every ROI. We score recall, the fraction of ROIs profiled. Gene sets and ROIs are computed from ground truth and used only for scoring.

## 3.4 Imputer

One imputer serves every policy, so differences between policies reflect selection alone. It maps image principal components through random Fourier features [34], fits a ridge regression to the profiled regions, and propagates the resulting predictions over a fixed 8-nearest-neighbour graph in image and spatial features while holding profiled regions at their measured values, as in harmonic label propagation [49]. This is a light analogue of neighbourhood-aware predictors such as STFlow [20], cheap enough to refit after every measurement. Its predictions depend on the measured values. Its uncertainty, one minus the graph influence that reaches a region from profiled nodes, depends only on which regions are profiled. A $k$-nearest-neighbour imputer and a regression-kriging imputer [18] were evaluated on an earlier slide set (Appendix C). Hyperparameters of the imputer and policies were fixed before the registered analysis in which each was first used and were not tuned on its outcomes; those of the imputer were set during development on one prostate slide (Appendix B).

## 3.5 Policies

*Non-adaptive designs.* Random; farthest-point selection in space (grid), a greedy $k$-centre cover [16]; $k$-means [27] on image embeddings, taking the region nearest each centroid; a reimplementation of

a one-shot image ranker [44]; and variance- and mutual-information (MI)-greedy placement [24] with an RBF kernel on image and coordinate features and a median-heuristic lengthscale [14].

*Sequential fixed designs.* Error-reduction [36] picks the region whose 15 nearest unprofiled neighbours in the imputer's feature space carry the most uncertainty, and maximum-uncertainty picks the most uncertain region [25]. Because the imputer's uncertainty does not depend on measured values, both are fixed designs computed one region at a time; for error-reduction we confirmed that permuted values leave its selections unchanged.

*Value-conditioned policies.* The *residual* policy keeps the error-reduction form but replaces the uncertainty with an estimate of the imputer's actual error. It computes the leave-one-out residual of the image ridge model at each profiled region from the hat matrix [17], averages it over each unprofiled region's 10 nearest profiled neighbours, and multiplies by one minus the graph influence. The *local-variance* policy, our detection policy, scores a candidate by the kernel-weighted variance of measured expression around it plus 0.5 times its distance to the nearest profiled region, an exploitation term for heterogeneity balanced by an exploration term as in active search [13].

*Competitor objectives.* A diversity sampler applies farthest-point selection in joint image and spatial features, the coverage criterion of core-set active learning [37], standing in for the coverage-dominated reward of SCR$^2$-ST [47]. An adaptive ROI-seeking sampler follows the reward of SOFisher [26]. Both reimplement stated objectives, not the authors' trained models.

*Hybrids* (RQ3). A hybrid takes its first $\lceil fn \rceil$ regions, for a budget of $n$ regions and $f \in \{0.5, 0.8\}$, from a fixed coverage design and the rest from a feedback policy: the residual policy for reconstruction and local-variance for detection. This mirrors the space-filling start of adaptive global optimisation [22]. Because the coverage designs select in a fixed order, the first phase is exactly the prefix of the design's own selection (verified for all four designs at both fractions), so a hybrid and its coverage design differ only in how the last $1 - f$ of the budget is spent.

### 3.6 Feedback ablations

Each value-conditioned policy and hybrid was also run with uninformative feedback while its selection rule was left unchanged. Under *permuted* feedback, profiling a region reveals the expression of a randomly chosen other region, so the residual policy's leave-one-out model, for example, is fitted to permuted values. Under *constant* feedback, every region appears to have the slide's mean expression. The imputer and the scores always use the true values. A policy that gains from feedback should outperform its own permuted version, the logic of permutation tests for whether a learner exploits real structure [32]. Stage 2 used one permutation; stage 3 used two per configuration and repeated the stage-2 ablations with up to four further seeds.

### 3.7 Comparators

The primary stage-2 comparator is a *feasible* design chosen without the target slide. For each held-out study, the non-adaptive design (grid, $k$-means, variance-greedy, MI-greedy, or one-shot) with the best mean score on the other 17 studies at that budget and objective is applied to the held-out slides, a leave-one-study-out form of cross-study validation [3]. The best design for each slide chosen in hindsight is reported only as a benchmark, since selecting on the evaluation data inflates the comparator [6, 41]. Stage 3 compared each hybrid with a single design fixed for all slides at each budget and objective: the design the feasible procedure chose for most held-out studies. Those designs were chosen on stage-2 results from the same slides, which favours the baseline.

### 3.8 Statistical analysis and registration

Each margin is a per-slide difference between two policies. We report its mean with a 95% percentile bootstrap interval [11] that resamples whole studies [5, 12] (5,000 resamples), and an exact two-sided sign-flip test over the $2^{18}$ study-level sign patterns [42]. Intervals are not adjusted for multiplicity; $p_H$ denotes $p$-values adjusted by Holm's procedure within a registered family [19]. To make null results interpretable [1], we report the minimum detectable effect [4]: the smallest true mean margin detected with 80% power [8], estimated by simulation that resamples studies around the observed dispersion [2]. For a family of $m$ margins we compute it at $\alpha = 0.05/m$ (MDE$_H$), a conservative bound for the Holm decision.

The sign-flip test assumes exchangeable study margins. This holds for a fixed comparator but not exactly for the feasible one, whose selection for each study uses the other 17. Following the principle that model selection must be repeated inside resampling [41], we checked the stage-2 results post hoc with a bootstrap that repeats the selection in every replicate and with a single fixed comparator (Appendix F).

Analyses were registered in the repository in three stages [31], each before its experiments and after the previous stage's results were known.[1] Stage 1 compared policies with hindsight benchmarks in a family of 24 margins, including the first batched experiments (Appendix D). Stage 2 introduced the feasible comparator, the residual policy, and the feedback ablations, with a primary family of 18 margins: residual minus feasible at six budgets (RQ1), and at four budgets local-variance minus its permuted version, minus the feasible design, and minus the diversity sampler (RQ2). Stage 3 tested the hybrids, with a primary family of 20 margins, each hybrid minus the same coverage design run for the whole budget (RQ3), and a secondary family of 24 that compared hybrids with their permuted versions and repeated the batched experiments with the fixed detection design and permuted controls (RQ4). In every stage a benefit was to be claimed only for a positive margin with $p_H < 0.05$. Results are labelled confirmatory (registered primary), secondary (registered, Holm within the secondary family), descriptive, or post hoc.

### 3.9 Positive controls

A null result is informative only if the protocol could have registered an effect. In synthetic slides of $20 \times 20$ regions with spatially clustered structure that the image cannot predict (8 slides per setting, 10% budget, $k$-NN imputer), we varied the correlation length of the hidden structure and the amplitude $\alpha$ with which ROIs appear in measured expression. On the 22 real slides we injected 20 genes carrying three spatial hotspots of radius 5% of the slide diagonal, in the manner of simulated spatially variable genes [40], with standard deviation $\alpha \in \{0, 1, 2\}$ times the median of the slide's top-50 genes. Because the image cannot predict the hotspots, only measured values reveal them. Before running this control we set its success criterion: the residual policy beats the feasible design with an interval excluding zero at $\alpha = 2$.

## 4 Results

### 4.1 Positive controls

The simulator registered an effect of feedback on detection. In synthetic slides, the local-variance policy's recall margin over coverage rose with ROI amplitude, from $-0.045$ at $\alpha = 0$ to $+0.052$ (95% CI $+0.033$ to $+0.071$) at $\alpha = 4$ and $+0.101$ ($+0.065$ to $+0.135$) at $\alpha = 8$, positive on all 8 slides at both amplitudes (Figure 1b). Reconstruction behaved differently: even an ordering by true local difficulty, which no policy could deploy, stayed below coverage at every correlation length, by $0.105$ and $0.066$ PCC at the two shortest and $0.013$ to $0.020$ at the longer ones (Figure 1a).

The semi-synthetic control failed its criterion. At $\alpha = 1$ and 2, all 20 injected genes entered the evaluation set on every slide, yet at $\alpha = 2$ the residual policy remained below the feasible design, by $0.021$ ($0.012$ to $0.030$) PCC at 5% and $0.011$ ($0.004$ to $0.019$) at 10%, close to its deficits without injection ($0.015$ and $0.007$; Figure 2c, Appendix H). Feedback still helped relative to permuted values, by $+0.001$ to $+0.013$. The protocol can thus register a large detection effect, but not demonstrably a reconstruction gain, and the RQ1 results below concern the tested policies.

### 4.2 RQ1: Reconstruction

No value-conditioned policy outperformed the feasible design (Tables 1 and 2, Figure 2a). The residual policy fell below it at every budget, by $0.092$ PCC at 0.5%, $0.057$ at 1%, $0.025$ at 2%, $0.016$ at 5%, $0.007$ at 10%, and $0.002$ at 20%, significantly so at 0.5, 1, 5, and 10% ($p_H \leq 0.010$). The local-variance policy was further below, by $0.119$ to $0.163$ (secondary, $p_H \leq 0.001$). Bootstrap intervals that repeat the comparator's selection in each replicate excluded zero at every budget (post hoc; Appendix F).

[1] The repository plans label RQ4 as Q3 and record stage 3 as iteration 11 of `docs/ITERATION_LOG.md`.

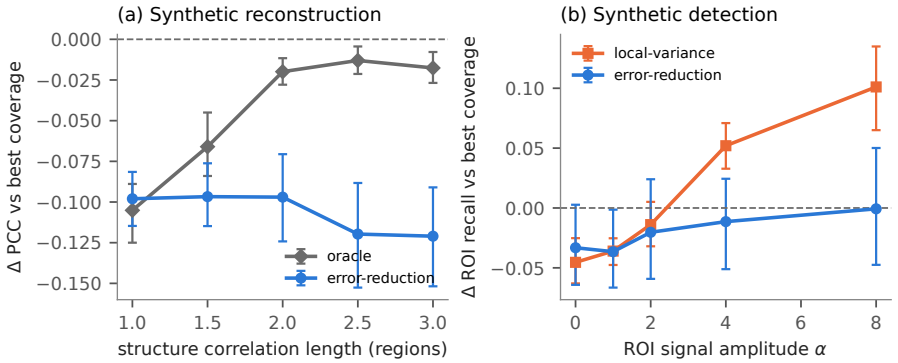



Figure 1: Synthetic controls (8 slides per point, 10% budget, mean and 95% bootstrap CI over slides). (a) Reconstruction margin over the best coverage design versus the correlation length of the hidden structure, for an ordering by true local difficulty and for error-reduction. (b) Detection-recall margin versus ROI amplitude.

Table 1: Reconstruction accuracy (PCC over the top-50 genes on unprofiled regions), mean over 22 slides. "Feasible" is the non-adaptive design chosen on the other studies; "per-slide best" is chosen in hindsight. "Residual, perm." is the residual policy given permuted feedback.

| | Non-adaptive | | Sequential fixed design | | Value-conditioned | | |
|---|---|---|---|---|---|---|---|
| Budget | feasible | per-slide best | error-red. | max-unc. | residual | residual, perm. | local-var. |
| 0.5% | 0.449 | 0.470 | 0.343 | 0.348 | 0.357 | 0.327 | 0.348 |
| 1% | 0.534 | 0.550 | 0.467 | 0.462 | 0.477 | 0.438 | 0.424 |
| 2% | 0.591 | 0.607 | 0.567 | 0.562 | 0.566 | 0.536 | 0.471 |
| 5% | 0.659 | 0.660 | 0.650 | 0.648 | 0.643 | 0.633 | 0.513 |
| 10% | 0.685 | 0.687 | 0.684 | 0.685 | 0.678 | 0.677 | 0.535 |
| 20% | 0.707 | 0.709 | 0.708 | 0.710 | 0.705 | 0.705 | 0.544 |

The measured values nonetheless carried information. With informative rather than permuted feedback, the residual policy gained 0.031, 0.039, and 0.030 PCC at 0.5–2% (secondary, Appendix G; $p_H$ 0.056, 0.002, 0.014), with no difference at 10–20%. Feedback moved the policy toward the feasible design without closing the gap. The error-reduction design, which shares the residual policy's structure but uses value-independent uncertainty, trailed the feasible design by similar amounts (0.107, 0.067, and 0.024 at 0.5–2%; post hoc), and both converged on it as the budget grew. The only design to exceed the feasible design was maximum-uncertainty, by 0.003 PCC at 20% (95% CI 0.002 to 0.004, $p = 0.0007$, post hoc), a fixed design that uses no feedback. The hindsight benchmark exceeded the feasible design by 0.016 to 0.021 PCC at budgets up to 2% and by at most 0.002 above (Appendix E).

### 4.3 RQ2: Detection

Informative feedback raised the local-variance policy's recall over permuted feedback by 0.005, 0.017, 0.029, and 0.063 at 2, 5, 10, and 20% (Table 3, Figure 2b). The unadjusted intervals excluded zero at 5–20%, but no margin survived adjustment ($p_H \geq 0.29$), and each estimate lay below its family-wise MDE (0.015, 0.035, 0.066, 0.112). The margin over the feasible design was smaller (0.006 to 0.026, $p_H \geq 0.25$), and the margin over the diversity sampler was close to zero (0.001 to 0.012, $p_H = 1.000$).

The absolute recalls locate the feedback effect (Table 4). At 20%, the policy recovered 0.278 of the ROIs with informative feedback but only 0.215 with permuted and 0.204 with constant values, below the feasible design (0.252); the diversity sampler reached 0.266 without feedback. Part of the feedback effect therefore recovers ground that the selection rule loses on its own. The margin over constant feedback at 20% was the only significant detection effect of feedback (+0.074, $p_H = 0.038$, secondary).

Table 2: Reconstruction margins (PCC), 22 slides, 18 studies. $\Delta$: mean per-slide difference with an unadjusted 95% CI resampling studies. $p_H$: Holm-adjusted within the stage-2 primary or secondary family; $p$: unadjusted (post hoc).

| | residual − feasible (primary) | | residual − residual, perm. (secondary) | | error-red. − feasible (post hoc) | |
|---|---|---|---|---|---|---|
| Budget | $\Delta$ [95% CI] | $p_H$ | $\Delta$ [95% CI] | $p_H$ | $\Delta$ [95% CI] | $p$ |
| 0.5% | $-0.092$ $[-0.132, -0.054]$ | 0.002 | $+0.031$ $[+0.010, +0.057]$ | 0.056 | $-0.107$ $[-0.149, -0.067]$ | <0.001 |
| 1% | $-0.057$ $[-0.089, -0.028]$ | 0.010 | $+0.039$ $[+0.021, +0.060]$ | 0.002 | $-0.067$ $[-0.093, -0.044]$ | <0.001 |
| 2% | $-0.025$ $[-0.041, -0.009]$ | 0.099 | $+0.030$ $[+0.014, +0.049]$ | 0.014 | $-0.024$ $[-0.039, -0.007]$ | 0.013 |
| 5% | $-0.016$ $[-0.025, -0.009]$ | 0.003 | $+0.009$ $[+0.001, +0.018]$ | 0.238 | $-0.010$ $[-0.017, -0.003]$ | 0.021 |
| 10% | $-0.007$ $[-0.011, -0.003]$ | 0.010 | $+0.001$ $[-0.002, +0.004]$ | 1.000 | $-0.001$ $[-0.006, +0.003]$ | 0.763 |
| 20% | $-0.002$ $[-0.004, -0.001]$ | 0.126 | $-0.001$ $[-0.003, +0.002]$ | 1.000 | $+0.001$ $[-0.002, +0.003]$ | 0.537 |

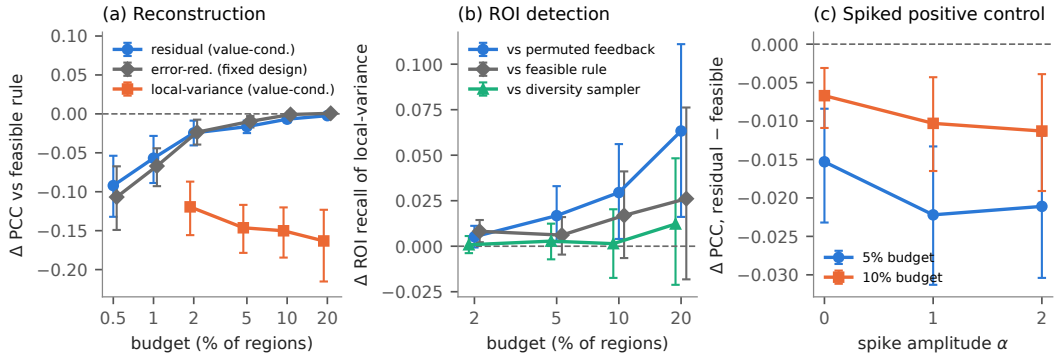



Figure 2: Stage-2 margins over 22 slides, mean and unadjusted 95% CI resampling studies. (a) Reconstruction margin over the feasible design. (b) Detection-recall margin of the local-variance policy over its permuted-feedback version, the feasible design, and the diversity sampler. (c) Spiked control: residual policy minus feasible design by spike amplitude.

The apparent advantage also depends on the comparator. Against MI-greedy fixed for every slide, the local-variance policy gained 0.043 and 0.078 recall at 10 and 20% ($p_H = 0.043$ for both; post hoc). MI-greedy, however, is a reconstruction design and the weakest detection design in the pool at 5–20% (0.200 recall at 20%, against 0.247 for variance-greedy and 0.252 for the one-shot ranker), and selection-aware intervals against the feasible design included zero at every budget (Appendix F). The ROI-seeking port did not improve on coverage in stage 1 (Appendix D). Because the port is not the trained SOFisher, we also ran SOFisher natively on its mouse cortex data, where its reward for captured target regions rose from a median of $-238$ early in training to $+147$ late (descriptive; Appendix L); the port's result is therefore not a test of SOFisher. SCR$^2$-ST could not be run natively because its processed training data are not public.

### 4.4 RQ3: Feedback on top of a coverage design

No hybrid outperformed its coverage design (Table 5, Figure 3). For reconstruction, the hybrids recovered most of the residual policy's deficit, but the half-coverage hybrid remained below $k$-means at 0.5%, by 0.036 PCC ($p_H = 0.006$), and from 2% upward every hybrid was within 0.007 PCC of MI-greedy with no significant margin. For detection, every hybrid margin lay between $-0.003$ and $+0.006$ recall ($p_H \geq 0.76$). These nulls are tight: at 2–10% the family-wise MDEs were 0.004 to 0.017 PCC and 0.004 to 0.026 recall, so gains above about 0.02 PCC or 0.03 recall would probably have been detected. At 20% the detection bound widens to 0.035–0.068.

Feedback within the hybrids was informative. With permuted values the hybrids did worse at 2%, by 0.016 and 0.007 PCC for the two reconstruction hybrids and 0.007 recall for the half-coverage detection hybrid (secondary, $p_H \leq 0.027$). For that detection hybrid the difference grew to 0.014–0.035 at 5–20% without surviving correction; for reconstruction at 10–20% the permuted versions were marginally ahead, by at most 0.003 PCC ($p_H \geq 0.12$; Appendix I). The absolute scores show what the information bought (Table 16). With permuted values the feedback phase lost ground

Table 3: Detection margins (ROI recall) of the local-variance policy, 22 slides. $\Delta$ with unadjusted 95% CI resampling studies; $p_H$ Holm-adjusted over the 18 stage-2 primary margins; $\text{MDE}_H$ at 80% power and $\alpha = 0.05/18$.

| | local-var. − permuted feedback | | | local-var. − feasible | | local-var. − diversity | |
|---|---|---|---|---|---|---|---|
| Budget | $\Delta$ [95% CI] | $p_H$ | $\text{MDE}_H$ | $\Delta$ [95% CI] | $p_H$ | $\Delta$ [95% CI] | $p_H$ |
| 2% | $+0.005$ $[-0.001, +0.011]$ | 1.000 | 0.015 | $+0.008$ $[+0.002, +0.014]$ | 0.250 | $+0.001$ $[-0.004, +0.006]$ | 1.000 |
| 5% | $+0.017$ $[+0.002, +0.033]$ | 0.530 | 0.035 | $+0.006$ $[-0.005, +0.016]$ | 1.000 | $+0.003$ $[-0.007, +0.012]$ | 1.000 |
| 10% | $+0.029$ $[+0.004, +0.056]$ | 0.530 | 0.066 | $+0.017$ $[-0.006, +0.041]$ | 1.000 | $+0.001$ $[-0.017, +0.020]$ | 1.000 |
| 20% | $+0.063$ $[+0.016, +0.111]$ | 0.294 | 0.112 | $+0.026$ $[-0.018, +0.076]$ | 1.000 | $+0.012$ $[-0.021, +0.048]$ | 1.000 |

Table 4: ROI-detection recall, mean over 22 slides. "Observed", "permuted", and "constant" denote the local-variance policy with informative, permuted, and constant feedback. The oracle profiles regions in order of true heterogeneity and bounds recall from above.

| | Non-adaptive | | | | Value-conditioned (local variance) | | | |
|---|---|---|---|---|---|---|---|---|
| Budget | feasible | $k$-means | MI-greedy | diversity | observed | permuted | constant | oracle |
| 2% | 0.018 | 0.018 | 0.018 | 0.026 | 0.027 | 0.021 | 0.022 | 0.100 |
| 5% | 0.064 | 0.050 | 0.046 | 0.067 | 0.070 | 0.053 | 0.051 | 0.250 |
| 10% | 0.120 | 0.110 | 0.094 | 0.136 | 0.137 | 0.107 | 0.103 | 0.500 |
| 20% | 0.252 | 0.232 | 0.200 | 0.266 | 0.278 | 0.215 | 0.204 | 0.999 |

relative to continuing the coverage design, for example from 0.127 to 0.110 recall at 10%, and real values restored the hybrid to about the level of the design (0.133). In detection, the local-variance policy run alone (0.137 at 10%, 0.278 at 20%) scored above both hybrids at every budget (descriptive).

### 4.5 RQ4: Batched and contiguous acquisition (exploratory)

In stage 1, at 10% budget, the local-variance policy's detection margins over the hindsight benchmark were $+0.021$, $+0.034$, and $+0.011$ for rounds of 1, 5, and 25 regions and $+0.024$ for contiguous blocks, none with $p_H < 0.05$ (Appendix D). With a single round the policy has no measurements and falls back to its starting rule, the regions nearest the slide centroid; the resulting margin of $-0.051$ ($p_H = 0.008$) reflects that rule rather than adaptivity. Reconstruction margins were negative in every setting.

In stage 3, rounds of 5 regions did not reduce recall: the policy captured 0.150 and 0.288 of ROIs at 10 and 20%, slightly more than when choosing one region at a time (0.137 and 0.278; Table 6). Its margins over the fixed detection design ($+0.023$, $+0.035$) and over permuted feedback ($+0.050$, $+0.084$) did not survive correction ($p_H \geq 0.11$).

### 4.6 Robustness

*Permutation seeds.* Repeating the stage-2 ablations with further seeds (descriptive; Appendix I) did not change any conclusion. Across seeds the mean margin varied with a standard deviation of at most 0.007 PCC and 0.013 recall. The seed-averaged detection margins ($+0.005$, $+0.017$, $+0.033$, $+0.074$ at 2–20%) stayed below their family-wise MDEs, and for reconstruction the registered seed gave the largest of three margins at 0.5% ($+0.031$, against $+0.024$ and $+0.016$).

*Evaluation genes.* Choosing the top-50 genes from the whole slide uses information a prospective study would lack. Choosing them from a random 15% reference subset changed the stage-1 reconstruction margins by at most 0.003, and scoring all filtered genes left them negative ($-0.033$ to $-0.079$; Appendix J).

*Platform.* In an exploratory split, stage-1 detection margins were larger on Visium HD than on Visium ($+0.048$ against $+0.008$ at 10%), but with five HD studies the smallest attainable $p$ is 0.0625, and platform is confounded with study (Appendix K).

Table 5: Stage-3 primary margins: hybrid minus the same fixed coverage design run for the whole budget, 22 slides. The hybrid takes the first 50% or 80% of regions from the design and the rest from the feedback policy. $\Delta$ with unadjusted 95% CI resampling studies; $p_H$ Holm-adjusted over the 20 primary margins; $\text{MDE}_H$ at 80% power and $\alpha = 0.05/20$.

| | | coverage for 50% of budget | | | coverage for 80% of budget | | |
|---|---|---|---|---|---|---|---|
| Budget | fixed rule | $\Delta$ [95% CI] | $p_H$ | $\text{MDE}_H$ | $\Delta$ [95% CI] | $p_H$ | $\text{MDE}_H$ |
| *Reconstruction (PCC); feedback: residual policy* | | | | | | | |
| 0.5% | $k$-means | $-0.036$ $[-0.057, -0.018]$ | 0.006 | 0.056 | $-0.009$ $[-0.021, +0.003]$ | 1.000 | 0.033 |
| 1% | $k$-means | $-0.010$ $[-0.020, +0.002]$ | 1.000 | 0.021 | $-0.009$ $[-0.019, -0.001]$ | 1.000 | 0.025 |
| 2% | MI-greedy | $+0.001$ $[-0.007, +0.011]$ | 1.000 | 0.017 | $+0.007$ $[+0.002, +0.013]$ | 0.359 | 0.013 |
| 5% | MI-greedy | $-0.003$ $[-0.007, +0.002]$ | 1.000 | 0.013 | $-0.001$ $[-0.004, +0.003]$ | 1.000 | 0.008 |
| 10% | MI-greedy | $0.000$ $[-0.002, +0.002]$ | 1.000 | 0.005 | $+0.001$ $[-0.001, +0.002]$ | 1.000 | 0.004 |
| 20% | MI-greedy | $-0.002$ $[-0.004, +0.000]$ | 1.000 | 0.005 | $-0.001$ $[-0.003, +0.001]$ | 1.000 | 0.006 |
| *Detection (ROI recall); feedback: local-variance policy* | | | | | | | |
| 2% | variance-greedy | $+0.003$ $[+0.000, +0.005]$ | 0.756 | 0.007 | $0.000$ $[-0.001, +0.002]$ | 1.000 | 0.004 |
| 5% | variance-greedy | $+0.002$ $[-0.007, +0.010]$ | 1.000 | 0.020 | $0.000$ $[-0.003, +0.003]$ | 1.000 | 0.008 |
| 10% | variance-greedy | $+0.006$ $[-0.005, +0.018]$ | 1.000 | 0.026 | $+0.001$ $[-0.005, +0.007]$ | 1.000 | 0.015 |
| 20% | one-shot | $0.000$ $[-0.033, +0.034]$ | 1.000 | 0.068 | $-0.003$ $[-0.019, +0.014]$ | 1.000 | 0.035 |

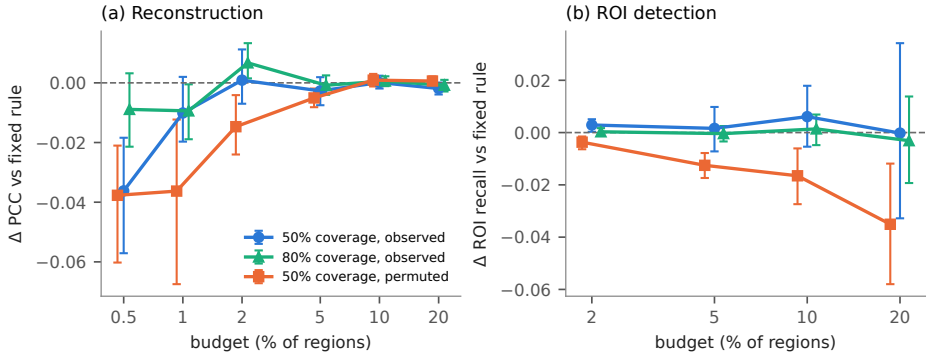



Figure 3: Hybrid minus the same fixed coverage design, 22 slides, mean and unadjusted 95% CI resampling studies. "Observed" hybrids receive real feedback (primary); the permuted series is the half-coverage hybrid given permuted feedback, averaged over two seeds (descriptive). Fixed designs as in Table 5.

## 5 Discussion

**Reconstruction.** The measured values were informative, yet using them did not help. Real feedback beat permuted feedback by up to 0.039 PCC when it drove the whole selection and by up to 0.026 when it chose only the remainder, but in both cases the gain mainly offset the cost of departing from a coverage design. This pattern is what sensor-placement theory leads one to expect when uncertainty depends on locations alone [24]. Our imputer's uncertainty has this property while its predictions do not, so any benefit of feedback had to come through the predictions. The evaluation does not test the theory, and the failed semi-synthetic control means we cannot exclude value-conditioned policies that would help; it shows that the obvious ones do not.

**Detection.** Here the answer depends on the baseline. Feedback-driven selection outperformed a design built for reconstruction (MI-greedy, post hoc) and, at the largest budget, its own constant-feedback version. Such gains are what active-search reasoning predicts [13], and the synthetic control confirmed that the simulator can show them. Against a design chosen for detection or a diversity sampler, the margins were small and uncertain, and giving the detection design a coverage phase within the policy did not help. The diversity sampler's near-parity suggests that, on these slides, heterogeneous regions are spread widely enough that broad coverage finds many of them. Resolving a difference of the size observed would need more independent studies than the 18 used here, especially at 20%.

Table 6: Batched acquisition, stage 3: local-variance policy revealing regions in rounds of 5, 22 slides. "One at a time" is the stage-2 sequential run. $\Delta$ with unadjusted 95% CI resampling studies; $p_H$ Holm-adjusted over the 24 stage-3 secondary margins; permuted values averaged over two seeds.

| | | ROI recall | | | | rounds of 5 − fixed rule | | rounds of 5 − permuted | |
|---|---|---|---|---|---|---|---|---|---|
| Budget | fixed rule | fixed | one at a time | rounds of 5 | rounds of 5, perm. | $\Delta$ [95% CI] | $p_H$ | $\Delta$ [95% CI] | $p_H$ |
| 10% | variance-greedy | 0.127 | 0.137 | 0.150 | 0.100 | $+0.023$ $[-0.000, +0.046]$ | 0.749 | $+0.050$ $[+0.022, +0.075]$ | 0.113 |
| 20% | one-shot | 0.252 | 0.278 | 0.288 | 0.204 | $+0.035$ $[-0.017, +0.095]$ | 1.000 | $+0.084$ $[+0.025, +0.141]$ | 0.240 |

**Implications.** For developers of adaptive acquisition, two comparisons should be standard: a fixed design chosen on other data for the same objective, and the method itself fed uninformative values. Neither comparison requires new data, and together they separate the value of feedback from the value of a good selection rule. Inference should treat studies, not slides, as the unit. For experimentalists, our evidence is confined to simulated spot-level acquisition at 0.5–20% of regions, one imputer, and computationally defined endpoints. Within those limits, no tested feedback policy outperformed a fixed design: $k$-means at the smallest budgets and MI-greedy above them for reconstruction, and variance-greedy, a one-shot ranker, or a diversity design for heterogeneity. Fixed designs also avoid the measurement turnaround that sequential selection requires. Whether this holds for ROI-scale acquisition, other predictors, or annotated biological targets would need prospective or annotated data. Our findings are consistent with the benchmark's result for predictor training [50], although the two studies address different decisions.

**Limitations.** The simulator is retrospective: a revealed region returns its measured expression without added noise, batch effects, or delay, conditions that if anything favour feedback. The ROIs are the 20% of regions with the highest local-variance score, not annotated structures. High local variance can mark tissue boundaries, mixtures of cell types, or technical noise as well as targets such as tumour margins or immune niches, so the detection results concern computational heterogeneity and do not show whether feedback helps find a particular biological target. The acquisition unit is a spot or HD pseudo-spot, smaller than GeoMx ROIs, and the two units differ in spacing. The batch and block simulations approximate but do not validate any platform. The 22-slide results use one imputer; the $k$-NN and kriging comparisons rest on five slides. The competitor samplers reimplement objectives; SOFisher was run natively only on its own data and SCR$^2$-ST not at all. The registered stage-2 ablation tests use one permutation, with further seeds descriptive. The feasible comparator, the residual policy, the ablations, and the hybrids were added after earlier results were known, although each was registered before its experiments. Budgets above 20% were not tested.

## 6 Conclusion

Across 22 human slides from 18 studies, conditioning region selection on measured expression did not improve whole-section reconstruction over a well-chosen fixed design, whether feedback drove the whole selection or only its remainder. For capturing heterogeneous regions, feedback helped relative to a design built for the other objective, but not detectably relative to a design built for detection. The value of feedback in region selection is therefore an empirical question for each objective, and it should be answered against feasible, objective-matched baselines and with uninformative-feedback controls.

**Reproducibility.** Code, configuration, result files, the registered analysis plans, and the scripts that regenerate every table and figure are available at `https://github.com/ayushdeo/STARE`.

## A Slides

Table 7 lists the slides. "Cluster" indexes the HEST study used as the resampling unit, and "Pitch" is the median nearest-neighbour spacing of regions.

Table 7: The 22 slides (HEST-1k v1.3 metadata).

| Tissue | HEST ID | Platform | Spots | $\mu$m/px | Pitch ($\mu$m) | Cluster | Study (HEST dataset title) |
|---|---|---|---|---|---|---|---|
| prostate | TENX174 | Visium HD | 2385 | 0.274 | 128 | 1 | Visium HD Spatial Gene Expression Library, Human Prostate ... |
| prostate | TENX40 | Visium | 4371 | 0.345 | 100 | 2 | Human Prostate Cancer, Adenocarcinoma with Invasive Carcin... |
| prostate | MEND160 | Visium | 4079 | 0.573 | 100 | 3 | Spatially resolved clonal copy number alterations in benig... |
| prostate | TENX46 | Visium | 3043 | 0.344 | 100 | 4 | Human Prostate Cancer, Acinar Cell Carcinoma (FFPE) |
| prostate | TENX41 | Visium | 2543 | 0.345 | 100 | 5 | Normal Human Prostate (FFPE) |
| prostate | MEND144 | Visium | 4074 | 0.341 | 100 | 3 | Spatially resolved clonal copy number alterations in benig... |
| kidney | TENX173 | Visium HD | 2684 | 0.274 | 128 | 6 | Visium HD Spatial Gene Expression Library, Human Kidney (F... |
| kidney | TENX71 | Visium | 5936 | 0.274 | 100 | 7 | Human Kidney, 11 mm Capture Area (FFPE) |
| kidney | INT21 | Visium | 4975 | 0.457 | 100 | 8 | Tertiary lymphoid structures generate and propagate anti-t... |
| kidney | INT19 | Visium | 4948 | 0.457 | 100 | 8 | Tertiary lymphoid structures generate and propagate anti-t... |
| lung | TENX171 | Visium HD | 1766 | 0.274 | 128 | 9 | Visium HD Spatial Gene Expression Libraries, Post-Xenium, ... |
| lung | TENX170 | Visium HD | 1877 | 0.274 | 128 | 9 | Visium HD Spatial Gene Expression Libraries, Post-Xenium, ... |
| lung | TENX163 | Visium HD | 2487 | 0.274 | 128 | 10 | Visium HD Spatial Gene Expression Library, Human Lung Canc... |
| lung | TENX72 | Visium | 6195 | 0.274 | 100 | 11 | Human Lung Cancer, 11 mm Capture Area (FFPE) |
| lung | TENX62 | Visium | 3858 | 0.273 | 100 | 12 | Human Lung Cancer (FFPE) |
| lung | MISC20 | Visium | 3316 | 0.438 | 100 | 13 | A spatially resolved atlas of the human lung characterizes... |
| colon | TENX156 | Visium HD | 2179 | 0.274 | 128 | 14 | Characterization of immune cell populations in the tumor m... |
| colon | TENX155 | Visium HD | 2281 | 0.274 | 128 | 14 | Characterization of immune cell populations in the tumor m... |
| colon | TENX70 | Visium | 9080 | 0.274 | 100 | 15 | Human Colorectal Cancer, 11 mm Capture Area (FFPE) |
| colon | TENX92 | Visium | 6518 | 0.274 | 100 | 16 | Visium CytAssist Gene Expression Libraries of Post-Xenium ... |
| colon | MISC47 | Visium | 4818 | 0.276 | 100 | 17 | COLON MAP: Colon Molecular Atlas Project |
| colon | ZEN48 | Visium | 2385 | 0.459 | 100 | 18 | Charting the Heterogeneity of Colorectal Cancer Consensus ... |

**Changes from an earlier seven-slide analysis.** An earlier analysis used seven Visium HD slides and described regions as 112 $\mu$m squares. The current HEST metadata lists two of those slides as non-human (TENX186, *Mus musculus*; TENX175, mixed human and mouse), and both are excluded here. The loader had also assumed 0.5 $\mu$m/px, whereas the HD images are about 0.274 $\mu$m/px, so those regions were about 61 $\mu$m squares containing one pseudo-spot each. On the five retained slides, per-slide local-variance margins (reconstruction and detection, four budgets) differ between the two analyses by a median of 0.015 (maximum 0.096); 3 of 40 change sign, two by less than 0.005 and one detection margin from $-0.050$ to $+0.015$. With seven slides the smallest attainable exact $p$-value is 0.016, so no earlier margin could have survived correction for 24 comparisons.

## B Configuration and implementation checks

All values are fixed in the released `config.py`. The imputer's hyperparameters were set during development on one prostate slide (TENX174, one of the 22) before any registered analysis and were not changed afterwards; the policies' hyperparameters were fixed before the stage in which each policy first appears.

Seeds: global 2021; random policy $\{0, \ldots, 4\}$; the stage-2 permutation uses the global seed, and stage 3 uses seeds 1 and 2 (1–4 for the repeated local-variance ablation). Imputer: image PCA 64, random Fourier features 512 (median-heuristic bandwidth), ridge 10, graph $k = 8$, 20 propagation steps, step size 0.5. $k$-NN imputer: $k = 10$, joint image and space features. Kriging: image PCA 64, ridge 10, residual $k = 10$. Placement kernels: feature PCA 64, median-heuristic lengthscale, noise $10^{-2}$. Error-reduction and residual policies: neighbourhood 15; the residual policy uses the 256 genes with the highest variance among profiled regions and 10 profiled neighbours. Local-variance: spatial kernel width 0.1 of the slide diagonal, exploration weight 0.5. Hybrids: coverage fraction $f \in \{0.5, 0.8\}$, rounded up. Batched stage-3 runs: rounds of 5. Spiked control: 20 genes, 3 hotspots, radius 0.05 of the diagonal, noise 0.1 of the reference standard deviation. Statistics: 5,000 bootstrap resamples; exact sign-flip enumeration over studies; MDE from 300 study resamples with 3,000 sign flips each, on a grid of 51 effect sizes.

The code for the larger slides replaces dense distance computations with sparse and incremental ones; a 132-run check (11 policies, 3 imputers, 2 slide sizes, 2 budgets) reproduced the original code's selections and scores exactly. In the stage-2 run, the five non-adaptive designs reproduced the stage-1 per-slide scores exactly, and the local-variance and diversity policies reproduced their stage-1 means. A caching error interrupted the first stage-2 run. For the results saved before it, a stale cache entry could only have produced a crash, and the error was fixed before any result was analysed.

## C Imputer comparison on the earlier slide set

The $k$-NN and kriging imputers were run only on the earlier slide set. Table 8 reports error-reduction minus the per-slide best non-adaptive design on the five human slides from that run, whose regions held one pseudo-spot each, with the number of slides on which the margin was positive. With five slides these values are descriptive.

Table 8: Error-reduction minus per-slide best non-adaptive design (PCC), mean over 5 human Visium HD slides (positive/total).

| Budget | kNN | kriging | graphflow |
|---|---|---|---|
| 0.5% | −0.088 (0/5) | −0.088 (2/5) | −0.082 (0/5) |
| 1% | −0.114 (0/5) | −0.138 (1/5) | −0.083 (0/5) |
| 2% | −0.091 (0/5) | −0.150 (1/5) | −0.040 (0/5) |
| 5% | −0.121 (0/5) | −0.141 (1/5) | −0.011 (1/5) |
| 10% | −0.134 (0/5) | −0.133 (0/5) | −0.000 (2/5) |
| 20% | −0.145 (0/5) | −0.126 (0/5) | −0.002 (2/5) |

## D Stage-1 results

Stage 1 compared policies with the per-slide best design (hindsight). Table 9 gives its 24-margin primary family except the batched and contiguous settings, which are in Table 10; Table 11 gives absolute recall, including the ROI-seeking sampler.

Table 9: Stage-1 primary margins (hindsight benchmarks). Error-reduction is compared with the per-slide best of five non-adaptive designs, local-variance with the per-slide best of $k$-means, grid, and MI-greedy. Unadjusted 95% CIs resampling studies; $p_H$ Holm-adjusted over the 24 stage-1 primary margins.

| | Reconstruction (PCC, top-50 genes) | | | | Detection (ROI recall) | |
|---|---|---|---|---|---|---|
| | error-reduction − best non-adapt. | | local-var. − best coverage | | local-var. − best coverage | |
| Budget | Δ [95% CI] | $p_H$ | Δ [95% CI] | $p_H$ | Δ [95% CI] | $p_H$ |
| 0.5% | −0.128 [−0.162, −0.096] | <0.001 | – | – | – | – |
| 1% | −0.083 [−0.105, −0.064] | <0.001 | – | – | – | – |
| 2% | −0.040 [−0.054, −0.027] | <0.001 | −0.136 [−0.171, −0.106] | <0.001 | +0.001 [−0.004, +0.006] | 0.723 |
| 5% | −0.011 [−0.018, −0.004] | 0.057 | −0.148 [−0.180, −0.119] | <0.001 | +0.013 [+0.002, +0.024] | 0.362 |
| 10% | −0.002 [−0.007, +0.001] | 0.687 | −0.152 [−0.186, −0.122] | <0.001 | +0.021 [+0.003, +0.037] | 0.362 |
| 20% | −0.002 [−0.004, +0.001] | 0.687 | −0.165 [−0.217, −0.125] | <0.001 | +0.042 [+0.006, +0.077] | 0.362 |

Table 10: Stage-1 local-variance policy minus the per-slide best coverage design (hindsight) at 10% budget under batched and contiguous acquisition. Δ with unadjusted 95% CI resampling studies; $p_H$ Holm-adjusted over the 24 stage-1 primary margins.

| | Reconstruction (PCC) | | Detection (ROI recall) | |
|---|---|---|---|---|
| Acquisition setting | Δ [95% CI] | $p_H$ | Δ [95% CI] | $p_H$ |
| Sequential (1 region per round) | −0.152 [−0.186, −0.122] | <0.001 | +0.021 [+0.003, +0.037] | 0.362 |
| Rounds of 5 regions | −0.129 [−0.175, −0.094] | <0.001 | +0.034 [+0.013, +0.053] | 0.073 |
| Rounds of 25 regions | −0.130 [−0.176, −0.095] | <0.001 | +0.011 [−0.000, +0.021] | 0.362 |
| Single round (whole budget) | −0.236 [−0.288, −0.188] | <0.001 | −0.051 [−0.077, −0.027] | 0.008 |
| Contiguous blocks of 7 regions | −0.087 [−0.116, −0.063] | <0.001 | +0.024 [+0.002, +0.044] | 0.362 |

Table 11: Stage-1 ROI-detection recall, mean over 22 slides. †Reimplementations of the stated objectives of SCR$^2$-ST (diversity, non-adaptive) and SOFisher (ROI-seeking, adaptive). *Ground-truth ordering.

| | Non-adaptive coverage | | | Competitor objectives | | | |
|---|---|---|---|---|---|---|---|
| Budget | $k$-means | grid | MI-greedy | diversity† | ROI-seeking† | local-var. | oracle* |
| 2% | 0.018 | 0.022 | 0.018 | 0.026 | 0.021 | 0.027 | 0.100 |
| 5% | 0.050 | 0.051 | 0.046 | 0.067 | 0.050 | 0.070 | 0.250 |
| 10% | 0.110 | 0.103 | 0.094 | 0.136 | 0.104 | 0.137 | 0.500 |
| 20% | 0.232 | 0.204 | 0.200 | 0.266 | 0.232 | 0.278 | 0.999 |

# E Feasible comparator choices and the hindsight gap

The feasible procedure selected $k$-means for reconstruction at 0.5–1% and MI-greedy at 2–20% (with $k$-means for 4 of 18 studies at 2%), and for detection variance-greedy for most studies at 2–10% and the one-shot ranker at 20% (Table 12).

Table 12: Designs chosen on the other studies (number of held-out studies, of 18) and the difference between the per-slide best design and the feasible design for reconstruction (unadjusted 95% CI resampling studies).

| Budget | feasible choice, reconstruction | feasible choice, detection | per-slide best − feasible (PCC) |
|---|---|---|---|
| 0.5% | kmeans (18) | – | $+0.021$ $[+0.006, +0.038]$ |
| 1% | kmeans (18) | – | $+0.016$ $[+0.006, +0.027]$ |
| 2% | mi-greedy (14), kmeans (4) | var-greedy (12), oneshot (5), grid (1) | $+0.017$ $[+0.008, +0.027]$ |
| 5% | mi-greedy (18) | var-greedy (18) | $+0.001$ $[+0.000, +0.003]$ |
| 10% | mi-greedy (18) | var-greedy (16), oneshot (2) | $+0.001$ $[+0.001, +0.003]$ |
| 20% | mi-greedy (18) | oneshot (18) | $+0.002$ $[+0.001, +0.003]$ |

# F Inference for the feasible comparator

The feasible comparator for each held-out study is chosen from the other 17, so the 18 study margins share the selection step, which the study-level tests in the main text do not account for. Two post hoc checks address this (Table 13). The selection-aware bootstrap resamples studies with replacement and repeats the leave-one-study-out choice inside each replicate, excluding copies of the held-out study from its own selection, so its interval reflects uncertainty in which design is chosen. The fixed-comparator analysis replaces the feasible design with MI-greedy on every slide, for which the sign-flip test is valid as specified, with Holm over the 18 analogous margins (the feedback and diversity margins do not involve the comparator and are unchanged). For reconstruction both checks agree with the main analysis. For detection the selection-aware intervals include zero at every budget, whereas the fixed MI-greedy margins exclude it at 10–20%; MI-greedy has the lowest detection recall in the pool at 5–20%.

Table 13: Stage-2 primary margins against the feasible design with selection-aware 95% bootstrap intervals, and against MI-greedy fixed for every slide (unadjusted 95% CI resampling studies, unadjusted and Holm-adjusted sign-flip $p$). Post hoc.

| | − feasible rule | | − MI-greedy (fixed) | | |
|---|---|---|---|---|---|
| Budget | $\Delta$ | selection-aware 95% CI | $\Delta$ [95% CI] | $p$ | $p_H$ |
| *Reconstruction (PCC): residual policy* | | | | | |
| 0.5% | $-0.092$ | $[-0.132, -0.020]$ | $-0.055$ $[-0.094, -0.001]$ | 0.057 | 0.424 |
| 1% | $-0.057$ | $[-0.087, -0.013]$ | $-0.045$ $[-0.072, -0.018]$ | 0.008 | 0.090 |
| 2% | $-0.025$ | $[-0.047, -0.011]$ | $-0.033$ $[-0.047, -0.017]$ | <0.001 | 0.014 |
| 5% | $-0.016$ | $[-0.025, -0.008]$ | $-0.016$ $[-0.025, -0.009]$ | <0.001 | 0.003 |
| 10% | $-0.007$ | $[-0.011, -0.003]$ | $-0.007$ $[-0.011, -0.003]$ | <0.001 | 0.011 |
| 20% | $-0.002$ | $[-0.004, -0.000]$ | $-0.002$ $[-0.004, -0.001]$ | 0.010 | 0.107 |
| *Detection (ROI recall): local-variance policy* | | | | | |
| 2% | $+0.008$ | $[-0.002, +0.013]$ | $+0.009$ $[+0.003, +0.015]$ | 0.017 | 0.168 |
| 5% | $+0.006$ | $[-0.004, +0.022]$ | $+0.023$ $[+0.010, +0.037]$ | 0.005 | 0.063 |
| 10% | $+0.017$ | $[-0.011, +0.041]$ | $+0.043$ $[+0.020, +0.067]$ | 0.003 | 0.043 |
| 20% | $+0.026$ | $[-0.018, +0.085]$ | $+0.078$ $[+0.036, +0.122]$ | 0.003 | 0.043 |

# G Stage-2 secondary family

Table 14: Stage-2 secondary margins, 22 slides. Unadjusted 95% CIs resampling studies; $p_H$ Holm-adjusted over the secondary family.

| Margin | Budget | $\Delta$ [95% CI] | $p_H$ |
|---|---|---|---|
| residual − residual, perm. (PCC) | 0.5% | $+0.031$ $[+0.010, +0.057]$ | 0.056 |
| residual − per-slide best (PCC) | 0.5% | $-0.113$ $[-0.146, -0.084]$ | <0.001 |
| residual − residual, perm. (PCC) | 1% | $+0.039$ $[+0.021, +0.060]$ | 0.002 |
| residual − per-slide best (PCC) | 1% | $-0.072$ $[-0.099, -0.051]$ | <0.001 |
| residual − residual, perm. (PCC) | 2% | $+0.030$ $[+0.014, +0.049]$ | 0.014 |
| residual − per-slide best (PCC) | 2% | $-0.041$ $[-0.056, -0.029]$ | <0.001 |
| local-var. − local-var., constant (recall) | 2% | $+0.004$ $[-0.001, +0.010]$ | 0.518 |
| local-var. − feasible (PCC) | 2% | $-0.119$ $[-0.156, -0.087]$ | <0.001 |
| residual − residual, perm. (PCC) | 5% | $+0.009$ $[+0.001, +0.018]$ | 0.238 |
| residual − per-slide best (PCC) | 5% | $-0.018$ $[-0.026, -0.011]$ | <0.001 |
| local-var. − local-var., constant (recall) | 5% | $+0.019$ $[+0.006, +0.032]$ | 0.083 |
| local-var. − feasible (PCC) | 5% | $-0.146$ $[-0.178, -0.117]$ | <0.001 |
| residual − residual, perm. (PCC) | 10% | $+0.001$ $[-0.002, +0.004]$ | 1.000 |
| residual − per-slide best (PCC) | 10% | $-0.008$ $[-0.012, -0.005]$ | <0.001 |
| local-var. − local-var., constant (recall) | 10% | $+0.034$ $[+0.011, +0.057]$ | 0.076 |
| local-var. − feasible (PCC) | 10% | $-0.150$ $[-0.185, -0.120]$ | <0.001 |
| residual − residual, perm. (PCC) | 20% | $-0.001$ $[-0.003, +0.002]$ | 1.000 |
| residual − per-slide best (PCC) | 20% | $-0.005$ $[-0.007, -0.003]$ | 0.003 |
| local-var. − local-var., constant (recall) | 20% | $+0.074$ $[+0.032, +0.118]$ | 0.038 |
| local-var. − feasible (PCC) | 20% | $-0.163$ $[-0.215, -0.123]$ | <0.001 |

## H Spiked positive control

Table 15: Semi-synthetic control on the 22 slides: residual policy (and its permuted-feedback version) minus the feasible design, by spike amplitude and budget (PCC; unadjusted 95% CIs resampling studies). The registered success criterion was a positive margin with a CI excluding zero at $\alpha = 2$.

| $\alpha$ | Budget | spiked genes in top 50 | residual − feasible | residual, perm. − feasible | residual − residual, perm. |
|---|---|---|---|---|---|
| 0 | 5% | 0 | $-0.015\ [-0.023, -0.008]$ | $-0.026\ [-0.034, -0.018]$ | $+0.010\ [+0.002, +0.019]$ |
| 0 | 10% | 0 | $-0.007\ [-0.011, -0.003]$ | $-0.008\ [-0.012, -0.003]$ | $+0.001\ [-0.002, +0.004]$ |
| 1 | 5% | 20 | $-0.022\ [-0.031, -0.013]$ | $-0.030\ [-0.038, -0.023]$ | $+0.007\ [-0.001, +0.015]$ |
| 1 | 10% | 20 | $-0.010\ [-0.017, -0.004]$ | $-0.013\ [-0.019, -0.007]$ | $+0.003\ [-0.002, +0.007]$ |
| 2 | 5% | 20 | $-0.021\ [-0.030, -0.012]$ | $-0.034\ [-0.043, -0.028]$ | $+0.013\ [+0.004, +0.023]$ |
| 2 | 10% | 20 | $-0.011\ [-0.019, -0.004]$ | $-0.018\ [-0.026, -0.010]$ | $+0.007\ [-0.000, +0.014]$ |

## I Stage-3 absolute values, secondary family, and permutation seeds

Table 16: Stage-3 mean scores over 22 slides: the fixed coverage design, the hybrids with observed and permuted feedback (two seeds averaged), and the feedback policy run alone for the whole budget (residual policy for reconstruction, local-variance for detection; stage-2 runs). Fixed designs as in Table 5.

| | | hybrid, 50% coverage | | hybrid, 80% coverage | | |
|---|---|---|---|---|---|---|
| Budget | fixed rule | observed | permuted | observed | permuted | feedback policy alone |
| *Reconstruction (PCC)* | | | | | | |
| 0.5% | 0.449 | 0.413 | 0.412 | 0.441 | 0.437 | 0.357 |
| 1% | 0.534 | 0.524 | 0.498 | 0.525 | 0.517 | 0.477 |
| 2% | 0.599 | 0.600 | 0.584 | 0.605 | 0.599 | 0.566 |
| 5% | 0.659 | 0.656 | 0.654 | 0.658 | 0.657 | 0.643 |
| 10% | 0.685 | 0.685 | 0.686 | 0.686 | 0.687 | 0.678 |
| 20% | 0.707 | 0.705 | 0.708 | 0.706 | 0.708 | 0.705 |
| *Detection (ROI recall)* | | | | | | |
| 2% | 0.023 | 0.026 | 0.019 | 0.024 | 0.022 | 0.027 |
| 5% | 0.064 | 0.065 | 0.051 | 0.063 | 0.060 | 0.070 |
| 10% | 0.127 | 0.133 | 0.110 | 0.128 | 0.121 | 0.137 |
| 20% | 0.252 | 0.252 | 0.217 | 0.249 | 0.232 | 0.278 |

Table 17: Stage-3 secondary margins, 22 slides. Unadjusted 95% CIs resampling studies; $p_H$ Holm-adjusted over the 24 secondary margins. Permuted values are averaged over two seeds.

| Margin | Budget | $\Delta$ [95% CI] | $p_H$ |
|---|---|---|---|
| reconstruction, hybrid (50% coverage) − permuted | 0.5% | $+0.0015$ $[-0.0152, +0.0155]$ | 1.000 |
| reconstruction, hybrid (80% coverage) − permuted | 0.5% | $+0.0040$ $[-0.0012, +0.0100]$ | 1.000 |
| reconstruction, hybrid (50% coverage) − permuted | 1% | $+0.0262$ $[+0.0065, +0.0540]$ | 0.209 |
| reconstruction, hybrid (80% coverage) − permuted | 1% | $+0.0078$ $[-0.0002, +0.0161]$ | 0.749 |
| reconstruction, hybrid (50% coverage) − permuted | 2% | $+0.0156$ $[+0.0074, +0.0254]$ | 0.027 |
| detection, hybrid (50% coverage) − permuted | 2% | $+0.0066$ $[+0.0039, +0.0093]$ | 0.004 |
| reconstruction, hybrid (80% coverage) − permuted | 2% | $+0.0067$ $[+0.0040, +0.0094]$ | 0.016 |
| detection, hybrid (80% coverage) − permuted | 2% | $+0.0016$ $[+0.0002, +0.0028]$ | 0.477 |
| reconstruction, hybrid (50% coverage) − permuted | 5% | $+0.0024$ $[-0.0013, +0.0062]$ | 1.000 |
| detection, hybrid (50% coverage) − permuted | 5% | $+0.0141$ $[+0.0059, +0.0215]$ | 0.113 |
| reconstruction, hybrid (80% coverage) − permuted | 5% | $+0.0012$ $[-0.0012, +0.0039]$ | 1.000 |
| detection, hybrid (80% coverage) − permuted | 5% | $+0.0037$ $[+0.0011, +0.0059]$ | 0.277 |
| reconstruction, hybrid (50% coverage) − permuted | 10% | $-0.0008$ $[-0.0027, +0.0010]$ | 1.000 |
| detection, hybrid (50% coverage) − permuted | 10% | $+0.0227$ $[+0.0077, +0.0352]$ | 0.209 |
| reconstruction, hybrid (80% coverage) − permuted | 10% | $-0.0018$ $[-0.0031, -0.0006]$ | 0.124 |
| detection, hybrid (80% coverage) − permuted | 10% | $+0.0074$ $[-0.0004, +0.0139]$ | 0.749 |
| reconstruction, hybrid (50% coverage) − permuted | 20% | $-0.0025$ $[-0.0044, -0.0006]$ | 0.277 |
| detection, hybrid (50% coverage) − permuted | 20% | $+0.0349$ $[+0.0097, +0.0570]$ | 0.277 |
| reconstruction, hybrid (80% coverage) − permuted | 20% | $-0.0019$ $[-0.0039, -0.0002]$ | 0.477 |
| detection, hybrid (80% coverage) − permuted | 20% | $+0.0169$ $[+0.0041, +0.0291]$ | 0.277 |
| rounds of 5: local-variance − variance-greedy | 10% | $+0.0233$ $[-0.0001, +0.0463]$ | 0.749 |
| rounds of 5: local-variance − permuted | 10% | $+0.0497$ $[+0.0216, +0.0747]$ | 0.113 |
| rounds of 5: local-variance − one-shot | 20% | $+0.0352$ $[-0.0168, +0.0952]$ | 1.000 |
| rounds of 5: local-variance − permuted | 20% | $+0.0839$ $[+0.0247, +0.1405]$ | 0.240 |

Table 18: Stage-2 feedback ablations repeated over permutation seeds, 22 slides (descriptive). Seed 2021 is the registered stage-2 run. SD is across the seed means; the seed-averaged margin uses the mean of the permuted runs, with an unadjusted 95% CI resampling studies.

| | mean margin by permutation seed | | | | | | |
|---|---|---|---|---|---|---|---|
| Budget | 2021 (stage 2) | 1 | 2 | 3 | 4 | SD | seed-averaged $\Delta$ [95% CI] |
| *Reconstruction (PCC): residual − permuted* | | | | | | | |
| 0.5% | $+0.031$ | $+0.024$ | $+0.016$ | – | – | 0.007 | $+0.023$ $[+0.008, +0.042]$ |
| 1% | $+0.039$ | $+0.040$ | $+0.030$ | – | – | 0.006 | $+0.036$ $[+0.020, +0.055]$ |
| 2% | $+0.030$ | $+0.023$ | $+0.025$ | – | – | 0.004 | $+0.026$ $[+0.013, +0.041]$ |
| 5% | $+0.009$ | $+0.010$ | $+0.012$ | – | – | 0.002 | $+0.010$ $[+0.002, +0.019]$ |
| 10% | $+0.001$ | 0.000 | $+0.003$ | – | – | 0.002 | $+0.001$ $[-0.002, +0.005]$ |
| 20% | $-0.001$ | $-0.001$ | $+0.001$ | – | – | 0.001 | 0.000 $[-0.002, +0.001]$ |
| *Detection (ROI recall): local-variance − permuted* | | | | | | | |
| 2% | $+0.005$ | $+0.003$ | $+0.003$ | $+0.006$ | $+0.007$ | 0.002 | $+0.005$ $[-0.000, +0.010]$ |
| 5% | $+0.017$ | $+0.011$ | $+0.016$ | $+0.020$ | $+0.022$ | 0.004 | $+0.017$ $[+0.005, +0.030]$ |
| 10% | $+0.029$ | $+0.025$ | $+0.034$ | $+0.035$ | $+0.042$ | 0.006 | $+0.033$ $[+0.012, +0.056]$ |
| 20% | $+0.063$ | $+0.075$ | $+0.083$ | $+0.060$ | $+0.090$ | 0.013 | $+0.074$ $[+0.031, +0.118]$ |

# J Gene-set sensitivity

Table 19: Stage-1 local-variance minus best coverage reconstruction margin (PCC) under three gene sets, 22 slides, unadjusted 95% CIs resampling studies (secondary).

| Budget | per-slide top-50 HVG | reference-subset top-50 HVG | all filtered genes |
|---|---|---|---|
| 2% | $-0.136$ $[-0.171, -0.106]$ | $-0.134$ $[-0.168, -0.103]$ | $-0.032$ $[-0.047, -0.021]$ |
| 5% | $-0.148$ $[-0.180, -0.119]$ | $-0.148$ $[-0.182, -0.118]$ | $-0.052$ $[-0.068, -0.040]$ |
| 10% | $-0.152$ $[-0.186, -0.122]$ | $-0.153$ $[-0.189, -0.122]$ | $-0.068$ $[-0.087, -0.052]$ |
| 20% | $-0.165$ $[-0.217, -0.125]$ | $-0.166$ $[-0.217, -0.126]$ | $-0.079$ $[-0.101, -0.061]$ |

## K Platform and tissue splits

The platform split is exploratory and the per-tissue means are descriptive; both use stage-1 margins.

Table 20: Selected stage-1 margins by platform (unadjusted study-level sign-flip $p$) and by tissue (mean).

| | HD ($n$=7, 5 cl.) | Visium ($n$=15, 13 cl.) | prostate ($n$=6) | kidney ($n$=4) | lung ($n$=6) | colon ($n$=6) |
|---|---|---|---|---|---|---|
| Recon., error-red. (1%) | $-0.113$ (0.062) | $-0.068$ (<0.001) | $-0.078$ | $-0.085$ | $-0.081$ | $-0.088$ |
| Recon., local-var. (10%) | $-0.103$ (0.062) | $-0.174$ (<0.001) | $-0.115$ | $-0.171$ | $-0.141$ | $-0.185$ |
| Detect., local-var. (10%) | $+0.048$ (0.062) | $+0.008$ (0.426) | $-0.001$ | $+0.022$ | $+0.036$ | $+0.027$ |
| Detect., local-var. (20%) | $+0.099$ (0.062) | $+0.015$ (0.469) | $+0.015$ | $+0.024$ | $+0.052$ | $+0.070$ |
| Detect., rounds of 5 (10%) | $+0.049$ (0.250) | $+0.026$ (0.044) | $+0.016$ | $+0.012$ | $+0.046$ | $+0.053$ |

## L Native SOFisher run

We ran SOFisher from the authors' repository on its bundled mouse cortex data for 5 million steps (seed 222) with the repository's cortex script. The only change was checkpoint and resume support, so the run could continue after interruptions. Figure 4 shows the evaluation reward, which counts captured target regions. Its median was $-238$ over the first 5% of training and $+147$ over the last 10% (best 514). Rare evaluation episodes have rewards near $-1.7 \times 10^5$, so the figure shows rolling medians.

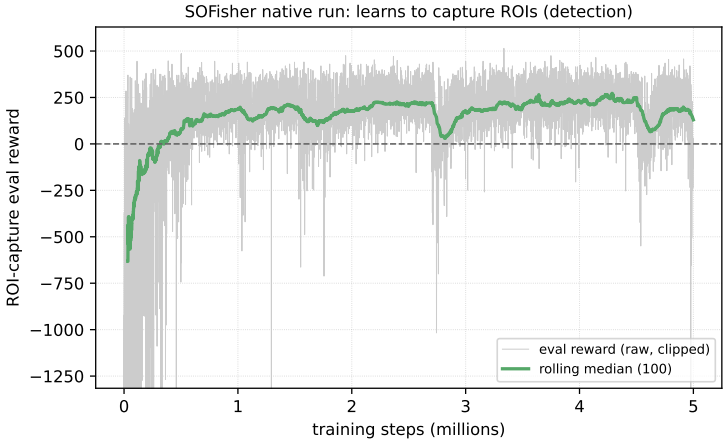



Figure 4: SOFisher evaluation reward during training (rolling median), native run.

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